MIL-P-89F
10 September 1982
SUPERSEDING
MIL-P-89E
6 June 1975

MILITARY SPECIFICATION

PAN, MESS KIT (CORROSION-RESISTING STEEL)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type and size of corrosion-resisting steel mess kit, pan.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-W-423	-	Wire, Steel, Corrosion-Resisting
PPP-B-636	-	Boxes, Shipping, Fiberboard

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7350

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MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

DRAWING

US ARMY NATICK RESEARCH AND DEVELOPMENT LABORATORIES

2-9-61 - Pan, Mess Kit, Corrosion-Resisting Steel

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet and Strip

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2 and 6.3).

3.2 Materials (see 6.4).

3.2.1 Steel, sheet and strip, corrosion-resisting. The corrosion-resisting steel sheet and strip shall conform to types 301, 302, or 304 of ASTM A 167.

3.2.2 Steel wire corrosion-resisting. Corrosion-resisting steel wire shall be annealed and shall conform to composition 302, or 304 of QQ-W-423.

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3.3 Construction. Construction of the mess kit pan shall conform to the design, dimensions, and tolerances shown on Drawing 2-9-61, and as specified herein. The mess kit body and cover shall be formed without seams from a single piece of corrosion-resisting steel specified in 3.2.1. All other components shall be fabricated from corrosion-resisting steel specified in 3.2.1 and 3.2.2, as applicable. Any riveting shall be accomplished so that the peening or upsetting operation is performed on the exterior surfaces of the body and cover. The hinge shall be attached by rivets or projection welds. The welds shall be tightly joined and of adequate size to satisfactorily comply with tests as specified in 4.5.2 and 4.5.3. Alignment of the pan mess kit shall be accomplished by mating the cover to the body. The body and cover shall be locked in the alignment position by placing the handle end over the body lip or edge.

3.4 Finish. The completed mess kit pan shall have a dull, frosted, lusterless finish equivalent to a commercial 2D finish on corrosion-resisting steel. When subjected to high humidity conditions, there shall be no evidence of rust, pitting, or spotting when tested as specified in 4.5.1.

3.5 Marking for identification. The letters US and the manufacturer's name or trademark of such known character, easily identifiable with said manufacturer, shall be permanently marked on the handle as shown on Drawing 2-9-61 and in accordance with MIL-STD-130. The surface opposite the markings shall show no signs of penetration or unevenness, nor shall the stamp marking cause distortion of any part of the handle.

3.6 Workmanship. The finished mess kit pan shall be clean and free from pits, scale, rust or discoloration and shall not contain any burrs, cracks, laps, splits, scrapes, slivers, sharp edges; rough die, tool, gouge, or grind marks. The body and cover shall not be galled by the forming operation or show any evidence of wrinkles, separation or blisters. There shall be no evidence of any component fractured, buckled, bent, dented, punctured, or malformed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for defects specified in 4.4.3.1.1 and 4.4.3.1.2 and tested for the characteristics specified in 4.4.3.2. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 In-process inspection. Examination of the welding process shall be performed prior to each days welding production by welding two (2) sample test specimens for each welding machine. Each welded test specimen shall be subjected to the peel test as specified in 4.5.2 and failure of any specimen shall be cause for correction to the welding process.

4.4.3 End item inspection. The inspection lot shall consist of all the mess kit pans offered for inspection at one time. The sample unit for this inspection shall be one completely assembled mess kit pan.

4.4.3.1 End item examination. The end item shall be examined in accordance with 4.4.3.1.1 and 4.4.3.1.2. The lot shall be expressed in units of mess kit pans. The sample unit shall be one mess kit pan assembly.

4.4.3.1.1 Visual examination. The complete mess kit pan assembly shall be examined for defects listed in table I. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not as specified		X
	Evidence of discoloration, rust, pits or stains	X	

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship (applicable to all components and assemblies)	Component missing	X	
	Improper fit of mating component, affecting proper closure	X	
	Body or cover not formed as one piece	X	
	Seams in body or cover	X	
	Component loose, mispositioned, not in proper alignment	X	
	Any galling in body or cover		X
	Any burrs, cracks, laps, splits, scrapes, slivers, wrinkles, or sharp edges, rough die, tool, gouge or grind marks	X	
	Any component fractured, buckled, bent, dented, punctured, or malformed	X	
Riveting	Rivets not peened or upset on exterior surface	X	
Welding	Contains crack or fracture	X	
	Not number specified	X	
	Not in proper location		X
	Contains burn hole, splatter or discoloration	X	
	Evidence of poor fusion	X	
	Any missing weld	X	
Identification marking	Missing, incomplete, illegible; not proper location		X
	Surface opposite marking shows penetration or unevenness		X

4.4.3.1.2 Dimensional examination. Examination shall be made of the sample unit to determine compliance with dimensions specified. Any deviation from specified requirements shall constitute a defect. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

4.4.3.2 End item testing. The end item shall be tested as specified in 4.5.1 and 4.5.3. The inspection level for the 4.5.1 test shall be S-2 with an AQL of 6.5 defects per hundred units. The inspection level for the 4.5.3 test shall be a sample of 3 mess kits per inspection lot and failure of any sample mess kit shall be cause for rejection of the lot.

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4.4.4 Packaging inspection. An examination shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged except that it need not be closed. Examination for closure defects listed below shall be made on shipping containers fully packaged. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged or not as specified
Workmanship	Inadequate application of components, such as: incomplete closure of container flap, improper taping, loose strapping, inadequate stapling Bulged or distorted container
Content	Number of pans per shipping container is more or less than specified

4.4.5 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 6.5 defects per hundred units.

Examine	Defect
Finished dimensions	Length, width or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Loads not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

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4.5 Methods of inspection.

4.5.1 Corrosion test. The completely assembled mess kit pan, shall be exposed to 95 to 100 percent relative humidity at room temperature for 16 hours to determine compliance with 3.4. Failure of any test shall be recorded as a defect.

4.5.2 In-process-peel test. The weld specimens shall be fabricated from the same material and thickness as used for the body and hinge. These pieces shall be welded together using the same projections, spacing and process as used for production welding. The welded specimens, two for each machine, shall be separated by driving a chisel between the welded components in an unwelded area or peeling one piece from the other until all welds fail. Failure outside the periphery of the weld area shall be considered evidence that the welds are satisfactory. Lack of fusion under the projection shall be cause for rejection of the item. It is not necessary to pull buttons in this test, provided that at least 3/4 of the area of each individual projection weld shows positive fusion. Failure to comply with these provisions shall be cause for rejection of the test sample.

4.5.3 End item weld test. The sample mess kit shall have the hinge separated from the body by driving a chisel between the components in an unwelded area or by peeling the hinge back until all welds fail. Failure outside the periphery of the weld area shall be considered evidence that the welds are satisfactory. Lack of fusion under the projection shall be cause for rejection. It is not necessary to pull buttons in this test, provided that at least 3/4 of the area of each individual projection weld shows positive fusion. Failure to comply with these provisions shall be cause for rejection of the item.

5. PACKAGING.

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each pan, assembled as shown in the illustration on Drawing 2-9-61, shall be placed in a 2-mil, minimum 8-1/2 inches by 14 inches, end-opening, commercial polyethylene bag. The open end of the bag shall be folded over the bottom of the mess kit pan and secured with pressure sensitive tape.

5.1.2 Commercial. Each pan shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Forty pans, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container, conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A unit packs shall be packed flat, four in length, two in width and five in depth within a shipping container. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

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5.2.2 Level B packing. Forty pans, preserved as specified in 5.1, shall be packed in a snug fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275, of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level A unit packs shall be packed flat, four in length, two in width and five in depth within a shipping container. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification, except the inspection shall be in accordance with 4.4.4.

5.2.3 Commercial packing. Mess kit pans, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Palletization. When specified (see 6.2), mess kit pans packed as specified in 5.2, shall be palletized in accordance with Load Type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit pack, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable.

6. NOTES

6.1 Intended use. Mess kit pans are intended for use by military personnel as a field mess plate and individual cooking utensil.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this document.
(b) When a first article sample is required (see 3.1, 4.3, and 6.3).
(c) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(e) When palletization is required (see 5.3).

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6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of one complete mess kit pan. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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MIL-P-89F Pan, Mess Kit (Corrosion-Resisting Steel)

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